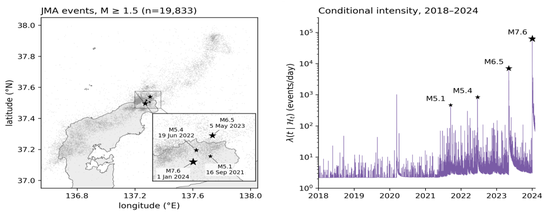
Distilling the E-Step for Fast Stochastic Declustering

Chijung Park, Wenbin Zhou, Shixiang (Woody) Zhu

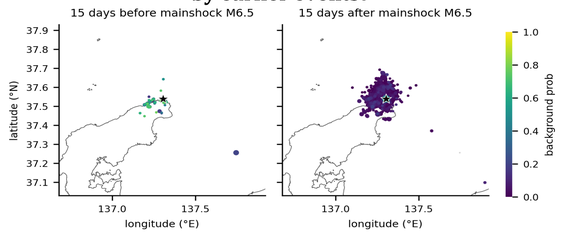
Carnegie Mellon University

I. Motivation

Noto Peninsula Earthquakes, 2018-2024



Which earthquakes are independent, and which are triggered by earlier events?



Estimated background probabilities near the 2023 Mainshock

II. Preliminaries

Self-exciting (Hawkes) processes: each event can temporarily raise the rate of future events.

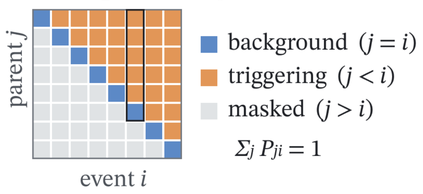
Conditional Intensity:

$$\lambda_\theta(x|\mathcal{H}_t)=\mu_\theta(x)+\sum_{x'\in\mathcal{H}_t} g_\theta(x,x')$$

Instantaneous Rate of event x given history $\mathcal{H}_t$

Stochastic declustering: Infers the probability that each event x_i is a background event or was triggered by an earlier event x_j :

Responsibility matrix P



Bottleneck of standard approach (EM):

- Repeats E- and M-step until convergence
- Recomputes $\mathcal{O}(n^2)$ responsibilities
- Refits from scratch for every new sequence

III. Proposed Method

Learn the E-step mapping from event sequences to responsibilities

$$\phi : \mathrm{D} \mapsto \hat{P}$$

Training: Student-Teacher

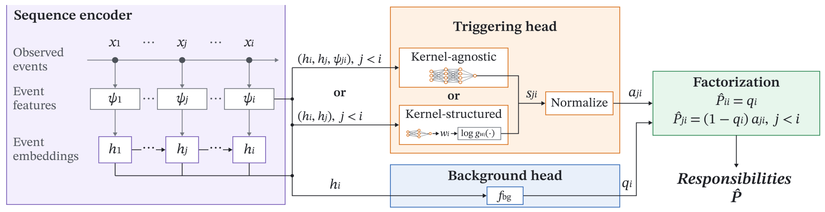
Teacher: Standard E-step on simulated sequences with known θ

Student: Neural network ϕ, trained by cross entropy:

$$\mathcal{L}(\phi) = -\frac{1}{n}\sum_{i=1}^{n}\sum_{j\le i} P^*_{ji}\log\hat{P}_{ji}$$

Neural Amortized Expectation Maximization (NAEM)

i) Neural E-step: Stochastic Declustering



Causal encoder → background & triggering heads → factorized P

ii) M-step: Parameter Estimation

$$\hat{\theta} = argmax_\theta Q(\theta|\hat{P}), \qquad Q(\theta|\hat{P})=\mathbb{E}_{Z\sim\hat{P}}[\log p_\theta(D,Z)]$$

Expected complete-data log-likelihood under neural responsibilities

IV. Theory

Computational Cost. For R iterations of EM,

$$\frac{Cost\ of\ EM}{Cost\ of\ NAEM} \approx \frac{R(\mathcal{T}_E+\mathcal{T}_M)}{\mathcal{T}_\phi+\tilde{\mathcal{T}}_M}$$

Error Theorem. Under local strong concavity(γ), oracle score rate, and approximate M-step stationarity(ε_{opt}) assumptions,

$$\|\hat{\theta}-\theta^*\| \le \frac{1}{\gamma}\left[\underbrace{\mathcal{O}_p\left(n^{-\frac{1}{2}}\right)}_{\text{finite-sample}} + \underbrace{\sqrt{V_n\epsilon_{KL}}}_{\text{amortization}} + \underbrace{\varepsilon_{opt}}_{\text{optimization}}\right]$$

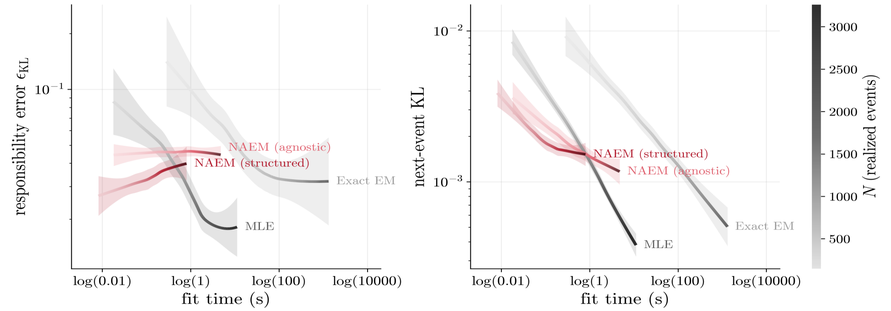
Score sensitivity (V_n): Sensitivity of the M-step score to responsibility error

Amortization Error (ϵ_{KL}): KL divergence between $\hat{P}$ *and* P^*

NAEM replaces R expensive EM iterations with one neural forward pass and a single M-step, introducing an amortization-error term $\sqrt{V_n\epsilon_{KL}}$, controlled by approximation quality of neural model ϕ

V. Results

Omori-Hawkes Simulation: *Is NAEM accurate, and how much faster?*

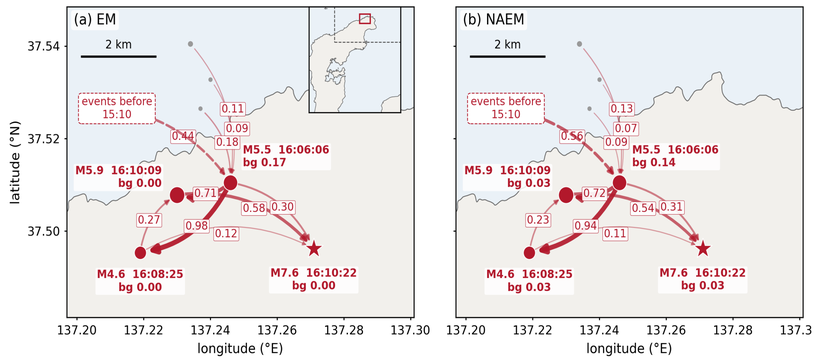


Left: Responsibility error. Right: Predictive accuracy (KL of next event distribution).

NAEM(red) achieves competitive accuracy with EM/MLE(gray) at a fraction of the fit time

Real-world Application: Noto Peninsula swarm (JMA catalog)

Inferred foreshock cascade: fitting 360 days before the M7.6 earthquake (9,106 events)



Arrows: parent → child, labeled with triggering probability P_{ji} ; bg: background probability P_{ii}

NAEM identifies same triggered events for M7.6 with close approximation of the posterior probability

Fit time of EM and NAEM for other windows

# of Events	EM	NAEM	Speedup
299 (30 d)	9.4 s	0.047 s	**201×**
958 (90 d)	50.4 s	0.45 s	**113×**
2,040 (180 d)	162 s†	2.1 s	**78×**
9,106 (360 d)	55 min	49 s	**67×**

† EM stopped at parameter bound (not converged)

NAEM approaches EM's accuracy with substantial acceleration